

Quantification of *Phytophthora palmivora* infection in barley and related monocot roots.

Matthew Macleod, Sabine Brumm, Sebastian Schornack

Sainsbury Laboratory, University of Cambridge, 47 Bateman Street, Cambridge, UK

matthew.macleod@slcu.cam.ac.uk; sebastian.schornack@slcu.cam.ac.uk

Abstract

Biotic stresses such as fungal pathogens significantly affect global crop yields. Understanding of the plant-pathogen interactions during root infection, especially in monocot crops, remains limited compared to fungal colonisations of dicots. The infection process of several cereal crop root-damaging fungi and oomycetes is highly similar to root infections by the pathogen model *Phytophthora palmivora*. Here we describe a protocol for quantifying root infection of *Phytophthora palmivora* in barley or related monocots using quantitative real-time PCR. This method can be used to produce relative expression values for the oomycete within roots based on RNA or DNA. Different mutations in the host can therefore be quantitatively assessed for their impact on infection. This allows us to identify plant processes underpinning susceptibility to filamentous fungal colonisation within monocot models and thus can facilitate the development of possible resistance strategies.

Keywords:

Phytophthora, *Phytophthora palmivora*, Barley, *Hordeum vulgare*, Monocot, Infection Assay, Oomycete, qPCR

1. Introduction

Pathogens and pests inflict significant global and regional losses to yields in food crop production. The top five most consumed crops globally can experience losses in yield between 15-30%, and higher losses in region-specific production hotspots [1]. As global climate changes continue to follow predicted trends, an increase in the susceptibility of economically important crops to different plant pathogens is expected [2], as well as the potential for the emergence of new strains of existing pathogens [3]. Amongst the threats to economically relevant crops, fungal pathogens are an important group and can affect various tissues in the plant. Examples such as rusts (*Puccinia* spp.), head blight (*Fusarium graminearum*), and powdery mildew (*Blumeria graminis*) can impact the monocot crops barley and wheat [1, 4, 5]. While these fungi have been studied infecting plant tissue above the ground, there has been less exploration of pathogenic fungi that target monocot roots. Therefore, it is important to develop methods to quantify root infection; such methods could be used for screening potential mutations that may decrease fungal colonisation and ultimately improve resistance in crops.

Filamentous pathogenic fungi can be challenging to culture reproducibly and achieving reliable infection of root tissues under laboratory conditions can be difficult. However, the filamentous “fungus-like” oomycete *Phytophthora palmivora* is easy to culture and sporulate and can be genetically transformed with fluorescent markers [5]. *P. palmivora* also reliably infects a broad range of plants, and infection is highly reproducible experimentally using spores, which can easily be set to a standard-concentration. Since oomycetes share similar mechanisms to filamentous fungi for invading host cells [6, 7], using *P. palmivora* as a model for this process has a direct contribution to understanding the molecular basis for invasion by filamentous pathogens, and possible strategies to reduce it. Our lab and others have studied *P. palmivora* infection in liverworts [8], legumes [9], barley [5], cocoa [10], oil palm [11] and rubber trees [12]. Experimental studies on *P. palmivora* across monocots and dicots have aimed to further characterise its infection mechanism in both leaf and root tissues [9, 13].

Here we describe a protocol for quantifying infection in *Hordeum vulgare* (barley) (and related monocot) roots by *P. palmivora*. The infection is performed on the roots of seedlings grown on agar plates, and levels of *Phytophthora* RNA are quantified via reverse transcriptase quantitative PCR (RT-qPCR) relative to transcript levels of host housekeeping genes. Normalisation of *P. palmivora* transcript levels to a barley housekeeping gene will account for variations in sample preparation and standardise expression for that sample. We use two housekeeping genes for reference to increase the accuracy of this. Barley housekeeping genes were chosen as a baseline, due to their consistent expression levels unaffected by infection. Nucleic acid-based quantification of pathogen biomass can be done directly using extracted DNA, although this captures dead material and may not be sensitive enough to detect low levels of pathogen during early infection. Conversely, reverse transcriptase RNA-based quantification captures the transcriptional profile of the infection, and so markers for highly expressed genes can be used to detect even low levels of pathogen. RT-qPCR starts from a reverse-transcribed DNA copy of total RNA and uses sequence-specific DNA-oligonucleotide primers to amplify a specific region of a target transcript, during which double-stranded DNA of this target accumulates, making this an efficient method for measuring nucleic acid levels with specificity at a time-point [14]. Calculating ratio values for a *P. palmivora* marker transcript relative to housekeeping transcripts allows for the comparison of infection levels between samples or replicates. This relative expression can help visualize the quantitative changes in *P. palmivora* infection across different barley or other monocot varieties and genotypes. Due to the reproducibility of this quantification, it is useful for surveying the germplasm and screening potentially advantageous mutations related to susceptibility to *P. palmivora*.

2. Materials and equipment

1. Sterile water
2. Ethanol (70%)
3. Sodium hypochlorite (Bleach)
4. Rotator Mixer (e.g. StarLab)
5. Medium-size plastic plates (e.g. Thermo Scientific™ Nunc™ Lab-Tek Petri Dishes, 79 sq. cm)
6. Whatman paper (e.g. Whatman® 3MM Chr paper 0.34 mm)
7. Forceps
8. Micropore tape
9. 24 °C growth cabinet
10. V8 Vegetable Media For 1 L: 100 mL V8 Juice (V8 brand), 1g calcium carbonate, 0.05 g β -sitosterol, 15 g Bacto-agar, 900 mL deionised water. Add carbenicillin to remove bacterial contamination (25mg/L).
11. Microbiological safety cabinet as required per local biosafety regulations for work with plant pathogens
12. Parafilm
13. Large-size plastic plates (e.g. Thermo Scientific™ 245 x 245 x 25 mm Nunc™ Square BioAssay Dishes)
14. 1% Water Agar (1g/100 mL agar in water)
15. 15mL centrifuge tubes
16. Miracloth (e.g. Merck Millipore Miracloth 22-25 μ m pore size)
17. Neubauer Haemocytometer
18. Transmission Light Microscope with 10x objective
19. Cellophane discs (e.g. 325P cellulose, A.A. Packaging Limited)
20. Black bin bag
21. Stereo/Dissecting Microscope
22. Flatbed Scanner
23. 50 mL centrifuge tubes
24. 2 mL microcentrifuge tubes (e.g. Eppendorf Safe-Lock Tubes)
25. 3mm tungsten beads for tissue lysis (e.g. Qiagen 3 mm Tungsten Carbide Beads)
26. Liquid Nitrogen
27. RNA extraction kit (e.g. Sigma-Aldrich Spectrum™ Plant Total RNA Kit)
28. Tissue Lyser (e.g. Qiagen TissueLyser II)
29. On-column DNase kit (e.g. Qiagen RNase-free DNase Set)
30. NanoDrop Spectrophotometer (e.g. Thermo Scientific™ NanoDrop™ One/One^C Microvolume UV-Vis Spectrophotometer)
31. cDNA synthesis kit (e.g. Roche Transcriptor First Strand cDNA Synthesis Kit)
32. Roche LightCycler® 480 SYBR Green I Master Mix
33. 384-well qPCR plate (e.g. LightCycler® 480 Multiwell Plate 384, white)
34. Clear qPCR plate seal (e.g. LightCycler® 480 Sealing Foil)
35. qPCR machine (e.g. Roche LightCycler® 480)

3.0 Methods:

3.1 Plant growth and preparation

This assay tests *Phytophthora palmivora* infection in isolation, and so plants are germinated under sterile conditions to prevent contaminant microorganisms from growing on the nutrient-rich agar medium the plants are moved onto. Sterilisation methods will vary depending on the plant system used. Here we will describe a protocol applicable to barley seeds.

1. Collect dried barley seeds in 50 mL centrifuge tubes (~ 40 seeds per tube) and add 40 mL 70% ethanol. Gently invert tubes manually or on a rotating machine (e.g. StarLab Rotator Mixer) for 5 min to ensure all seeds are covered.
2. Pour out the ethanol and wash once with 40 mL of sterile water. Gently invert for 1 min before pouring out.
3. Working in a lateral flow hood, add 40 mL of 4% sodium hypochlorite (bleach) and invert tubes for 10 min, and pour the liquid out. This time can be adjusted to combat any contamination on the seeds (see **Note 1**).
4. Perform three further washes with 40 mL of sterile water.
5. Prepare plastic plates (e.g. Thermo Scientific Nunc Lab-Tek Petri Dishes 79 sq. cm or similar) with three pieces of Whatman paper soaked in sterile water (enough so that a small pool forms when the plate is tilted).
6. Use sterile forceps to distribute seeds across the wet paper.
7. Seal plates tightly with Micropore tape and wrap the plates in tinfoil to block light.
8. Place plates at 4 °C for at least 5 d, and then move to 24 °C in a growth cabinet, still in the dark, for 3 d to germinate. See **Note 2** for issues with germination.

3.2 Maintenance of *P. palmivora* cultures

1. Grow *Phytophthora palmivora* on Petri dishes containing V8 vegetable media at 25 °C in constant darkness, sealed with Parafilm (see **Note 3**). See **Note 4** for combatting bacterial contamination and for antibiotic selection of *P. palmivora*.
2. To induce sporulation, after 5-7 d growth place plates under constant light at 24 °C and remove Parafilm to allow the plates to dry for 2-3 d.

3.3 Preparation of infection plates

1. Sterilise large plastic plates (e.g. Thermo Fisher 245x245x25mm) with 70% ethanol in a microbiological safety cabinet.
2. Pour 1% water agar into these plates (250 mL per plate or until half filled). Ensure that the plates have not dried excessively, as this will hinder plant growth.
3. With sterile forceps, place barley seedlings onto the middle line of the plate from left to right (see **Note 5**). It is possible to fit up to 20 seedlings per plate; we have performed assays comparing 5, 8, and 10 replicates of genotypes side-by-side on each half of the plate (**Note 6**).

3.4 Sporulation of *P. palmivora* for infection

1. Take the dried *P. palmivora* plates from 3.2, flood plates with 5 mL of sterile water for 20-30 min to harvest spores from the cultures (see **Note 7**).

2. Transfer the spore solution from the plates into a 15 mL centrifuge tube through sterile Miracloth.
3. Pipette 10 μ L of spore solution from the plate onto a Neubauer haemocytometer chamber. Count the spores with a transmission light microscope. Oomycete zoospores should be motile, unlike fungal spores.

Calculate the initial concentration of spores present per mL, and then dilute the spore solution to the desired concentration in sterile water (for more guidance, see **Note 8**).

3.5 Root inoculation with *Phytophthora* spores

1. Evenly pipette 100-150 μ L of the spore solution (or sterile water as a control, i.e. mock sample) over each individual plant's roots. This volume can be adjusted to achieve the desired total number of spores present on the plate.
2. Place cellophane disks with forceps carefully over roots (around 1-2 cellophane disks per 5 seedling roots) to cover them and trap *Phytophthora* spore solution (or water) as in **Figure 1**. This ensures even distribution of liquid over all parts of each root, as well as between roots of different seedlings (see **Note 9**).
3. Seal the edges of the plate with micropore tape.
4. Wrap the bottom of the plate so that the seed and roots are excluded from light (e.g. using aluminium foil or a black bin bag), but the shoots are not, and secure with tape. This simulates the roots being under soil and reduces potential stress and browning induced by light.
5. Leave plates lying flat under constant light at 24 °C. After 1 d, place plates vertically at a 100-110° angle (see **Note 10**).



Figure 1: Arrangement of Barley seedlings on 1% water agar under cellophane discs immediately after inoculation with *Phytophthora palmivora* spores. The *P. palmivora* strain being used is FLIMA (P7545). Three genotypes are being compared (A, B, C) on the same plate with six seedlings per genotype.

3.6 Tissue harvesting

The number of days allowed for infection depends on the species and strain of *Phytophthora* used and the stage of infection being investigated. Browning symptoms are often visible on the roots at the point of harvesting; however, this merely indicates a strong infection and is not essential. The best method to determine the level of infection visually is under a stereo/dissecting microscope to look for the formation of infection structures.

1. Scan plates with a flatbed scanner to get clear overview images of the infection symptoms.
2. Open plates in a microbiological safety cabinet and discard cellophane disks. Using forceps and a scalpel, cut the root system from each plant and dip it into a 50 mL centrifuge tube filled with sterile water to wash the roots. This way, only plant and oomycete materials are collected and used for PCR reactions. Do the same for any mock samples treated with water.
3. Dry roots with a paper tissue.
4. Place each root system into an individual 2 mL microcentrifuge tube.
5. Add three tungsten beads to each microcentrifuge tube to ensure complete lysis later of the sturdy barley roots.
6. Flash-freeze each tube in liquid nitrogen and store at -80 °C until required for nucleic acid extraction. Wear protective goggles as the liquid nitrogen can pop the tubes open, which could eject the metal beads out.

3.7 Nucleic acid extraction

Nucleic acid extraction can be performed for RNA or DNA, however RNA is a more sensitive marker for infection when there is a lower biomass of *Phytophthora* in the roots, for example during early infection stages.

1. Carry out RNA extraction using the kit specifications. The Sigma-Aldrich Spectrum™ Plant Total RNA Kit works effectively (see **Note 11**). Ensure that samples are not allowed to thaw (see **Note 12**).
2. For the initial preparation, grind the roots in a TissueLyser at 30 Hz for 1 min, then turn the tubes to face the other orientation and grind at 30 Hz for another minute to ensure even grinding of all tissue (see **Note 13**).
3. The *Phytophthora* can be considered inactivated completely after the lysis buffer has been added to the ground samples.
4. Perform on-column DNase treatment with RNase-Free DNase (e.g. Qiagen), following the manufacturer's instructions.
5. Calculate the concentration of RNA extracted for each sample, for example with a NanoDrop.

3.8 cDNA synthesis

1. Use 1µg of RNA per sample to synthesise cDNA using a cDNA Synthesis Kit, such as the Roche Transcriptor First Strand cDNA Synthesis Kit, following the manufacturer's recommendations (see **Note 14**).

3.9 qPCR:

1. Use the following reaction mix: cDNA (2.5 µL) (see **Note 15**), SYBR Green (5 µL), Nuclease-free sterile water (0.5 µL), Forward primer (1 µL), Reverse primer (1 µL) (all primers at a concentration of 10 µM). Total volume = 10 µL. Roche LightCycler® 480 SYBR Green I contains a master mix of Taq polymerase and dye.

Different primers can be used for measuring the relative *P. palmivora* RNA quantity against Barley housekeeping gene RNA levels.

Primers which have been used by our group are listed in **Table 1**, and further primers for marker genes in *P. palmivora* have been described previously [5].

2. Primer efficiencies should be calculated for later use in data analysis (see **Note 16**). For each sample, three technical replicates with each primer are used, totalling twelve reactions per plant sample.
3. Pipette master mixes of each reaction into a 384 well qPCR plate, before adding the cDNA and sealing the plate with a clear film. Label the qPCR plate by well, to describe the primer pair, sample and replicate in them (see **Note 17**).
qPCR conditions are as follows: Initial Denaturation 95°C 5 Min, Denaturation 95 °C 20s, Annealing 60 °C 20 s, Extension 72 °C 20s (45× cycles).
4. A melting curve can then be calculated with 1 cycle at 95 °C for 5 s and 65°C for 1 min, before 5 continuous acquisitions up to 97 °C (see **Note 18**).
5. Sequences of primers that can be used are in **Table 1**. *Phytophthora palmivora* Elongation Factor 1α (EF1α) (MH760192.1) is a biomass marker, and *H. vulgare* EF1α (KP293846.1) and Cyclophilin (Cyc) (CV056520) are housekeeping genes.

3.10 Data interpretation:

1. On the LightCycler 480 software, on the analysis page, perform an Absolute Quantification (2nd Derivative Max) and calculate the values in the table. Export this table as a txt file to create a table of Ct values corresponding to each reaction. Use the $2^{-\Delta\Delta C_t}$ method [15] to calculate the expression of *Phytophthora palmivora* RNA relative to reference housekeeping genes in barley.
2. A boxplot can then be used to effectively visualise relative expression levels across genotypes, treatments, and replicates, and compared using an unpaired student's T-test.

Table 1

Primer sequences for comparison of *Phytophthora palmivora* RNA expression relative to Barley (*Hordeum vulgare*) gene expression

Primer Name	Sequence (5'- 3')
<i>Phytophthora palmivora</i> EF1 α Forward	5'-CAAGATCCCGTTCGTGCCTA-3'
<i>Phytophthora palmivora</i> EF1 α Reverse	5'-GCGTTCAGGTTGTCAAGAGC-3'
<i>Hordeum vulgare</i> EF1 α Forward	5'-ATGATTCCCACCAAGCCCAT-3'
<i>Hordeum vulgare</i> EF1 α Reverse	5'-ACACCAACAGCCACAGTTTGC-3'
<i>Hordeum vulgare</i> Cyc Forward	5'-TTGAGGACGAGATAAGGCCAG-3'
<i>Hordeum vulgare</i> Cyc Reverse	5'-GCGACTGACAAGGTGCAAGAG-3'

4. Notes

1. If fungal contamination occurs, for example under the seed coat of barley seeds, inverting tubes during the bleach step can be extended to 30 min. A longer period of bleaching has proven effective for more rigorous sterilisation when contamination is present under the seed coats.
2. If seeds do not germinate evenly in this time, extending the number of days at 4 °C can give more time for germination and synchronisation between seeds.
3. Alternating the transfer of *Phytophthora* between spores and plugs week by week ensures a consistent pressure for sporulation within the population. The use of fluorescently tagged *Phytophthora* strains makes imaging more straightforward on plant material [5]. Fluorescence can be checked regularly on subsequent plates to ensure that expression is still strong and that an axenic population is consistently growing.
4. One common issue is the occurrence of bacterial contamination on the plates, which hinders zoospore production and renders cultures unusable for plant infections. To eliminate contamination, carbenicillin (25mg/L) can be added to V8 media plates. *P. palmivora* can also be transformed with a gentamycin antibiotic-resistance selection marker as described previously [16].
5. Drag seedlings slightly up to ensure roots are in contact with agar and are straightened and pointing to the bottom of the plate (see **Figure 1**). This ensures maximal contact with the *Phytophthora* solution

once it is added over the roots and distributed over the agar. Arranging roots like this will also help them grow evenly downwards and in constant contact with the media. If the roots appear to be shrivelled after a few days on the plates even with a mock treatment, ensure that they are in contact with the agar.

6. Comparisons between infections in different plant genotypes should be performed on the same plates to ensure that growth conditions are identical.
7. Placing *Phytophthora palmivora* plates at 4 °C for 20-30 min prior to flooding with water may also aid in stressing them to stimulate sporulation. It is advisable to stimulate sporulation in additional spare plates of *Phytophthora* in case sporulation is sporadic or poor.
8. Adjust the spore concentration (spores/mL) based on the host plant being affected, the *Phytophthora* species and strain being used, and the stage of infection being investigated. We use *Phytophthora palmivora* accession FLIMA (P7545) and find a concentration of 50,000 spore/mL is suitable when studying earlier stages of *Phytophthora* infection (3dpi). Our spore concentration is calculated by counting the average number of spores in large size haemocytometer squares (x10000 for spores/mL). We advise against an excessively large spore concentration if comparing resistance between genotypes. The high concentration results in high expression of pathogen RNA across all plants, and so small differences in resistance between plant genotypes may not be resolved.
9. Air bubbles can often form under the cellophane, so it is useful to carefully push these outwards using forceps, so that the spore solution or water gets evenly distributed around the roots and all plant samples are covered.
10. Leaving plates flat for the first day helps to prevent the spore solution from running immediately down the plate and allows the spores to collect and set into the roots. Plates are then grown at a 100-110° angle which prevents the roots from growing into the agar. This makes harvesting an easier process without the need to remove additional agar residue.
11. With the Spectrum RNA kit, protocol A is used on Step 4 for RNA binding as recommended for root tissue. Only a single final elution step is carried out.
12. Keep at -80 °C or in liquid nitrogen until the lysis solution has been added. Dry ice can also be used to temporarily hold samples. For optimal RNA yields, minimise the time taken for grinding tissue and immediately returns samples to liquid nitrogen between grinding and the addition of the lysis solution to prevent thawing of the tissue and degradation of RNA.
13. Tissue grinding is recommended to be performed with a grinding pestle (e.g. a StarLab microtube pestle) according to the Sigma Spectrum RNA Kit specifications. However, due to the strength of barley roots, we recommend using a TissueLyser with tungsten beads to give a more rigorous grinding. Leaving samples in the TissueLyser for slightly longer than is recommended helps to guarantee all samples are equally ground to a fine powder with barley roots. Where some tissue samples may not be completely turned into powder, additional grinding can be done with a pestle. A poor RNA yield can be the result of incomplete grinding of the root tissue.
14. Synthesise cDNA following kit recommendations and using Procedure A with anchored-oligo(dT)₁₈ primers. Carry out the optional initial denaturation step.

15. We dilute the cDNA 10-fold before the qPCR reactions with our primers.
16. The efficiency of two primer pairs in a reaction should be between 90-110%. Primer efficiency can be calculated by performing a serial dilution of cDNA containing the target gene, starting from a concentration of 1 (e.g. 10 ng DNA) and incremental dilutions by a factor of 10 or 2 [17]. These reactions should also be performed in triplicate. Plot the average Ct values for each dilution against the log₁₀ values for each concentration in the serial dilution. Primer efficiency is calculated using the equation $E=10^{-(1/\text{slope})}$ ($\times 100$ for %) as done previously [17, 18].
17. We recommend labelling each well of the qPCR plate with the sample and primer combination on the LightCycler software. This saves time later in matching the Ct values in the qPCR output file to the reaction that they correspond to.
18. Melt curve analysis shows whether binding has occurred for a primer pair to a secondary amplicon during the reaction. A melt curve analysis should be done both with primer efficiency calculation and during sample testing, to confirm the specificity of primer-target binding. A single bell-shaped curve indicates that the primer pair has only bound to the target gene. However, a second, smaller bell-shaped curve could indicate that primer dimerization or non-target binding has occurred, and the reaction may need to be repeated in case contamination has occurred in one of the reagents. Mock-treated (water) root samples should not have any amplification by the *P. palmivora*-specific primers, and so the melt curve for these reactions will not have any peaks.

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